

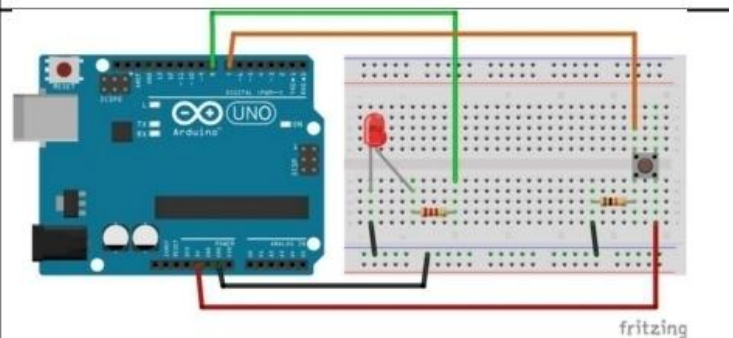
1. LED + PUSH BUTTON (ON/OFF)

```
#define LED_PIN 8
#define BUTTON_PIN 7

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT);
}

void loop() {
  if (digitalRead(BUTTON_PIN) == HIGH) {
    digitalWrite(LED_PIN, HIGH);
  } else {
    digitalWrite(LED_PIN, LOW);
  }
}
```

To study interfacing
LED and Push Button
with Arduino Board.



2. LED TOGGLE USING BUTTON

```
#define LED_PIN 8
#define BUTTON_PIN 7

byte lastButtonState = LOW;
byte ledState = LOW;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT);
}

void loop() {
  byte buttonState = digitalRead(BUTTON_PIN);
```

```

if (buttonState != lastButtonState) {
    lastButtonState = buttonState;

    if (buttonState == LOW) {
        ledState = (ledState == HIGH) ? LOW : HIGH;
        digitalWrite(LED_PIN, ledState);
    }
}
}

```

3. LED WITH DEBOUNCE

```

#define LED_PIN 8
#define BUTTON_PIN 7

byte lastButtonState = LOW;
byte ledState = LOW;

unsigned long debounceDuration = 50;
unsigned long lastTimeButtonStateChanged = 0;

void setup() {
    pinMode(LED_PIN, OUTPUT);
    pinMode(BUTTON_PIN, INPUT);
}

void loop() {
    if (millis() - lastTimeButtonStateChanged > debounceDuration) {

        byte buttonState = digitalRead(BUTTON_PIN);

        if (buttonState != lastButtonState) {
            lastTimeButtonStateChanged = millis();
            lastButtonState = buttonState;

            if (buttonState == LOW) {
                ledState = (ledState == HIGH) ? LOW : HIGH;
                digitalWrite(LED_PIN, ledState);
            }
        }
    }
}

```

3 LM35 TEMPERATURE SENSOR

```

float temp;
int tempPin = 0;

```

```

void setup() {
  Serial.begin(9600);
}

void loop() {
  temp = analogRead(tempPin);
  temp = temp * 0.48828125;

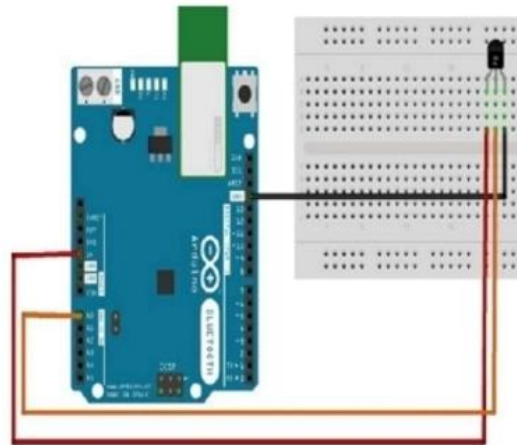
  Serial.print("TEMPERATURE = ");
  Serial.print(temp);
  Serial.print("*C");
  Serial.println();

  delay(1000);
}

```

To interface Temperature
Sensor LM35

LM35: Temperature



Connections:

Pin Connection

Left VCC (5V)

Middle Output → A0

Right GND

4. ULTRASONIC SENSOR

```

const int pingPin = 7;
const int echoPin = 6;

```

```

void setup() {
  Serial.begin(9600);
}

```

```

void loop() {
  long duration, inches, cm;

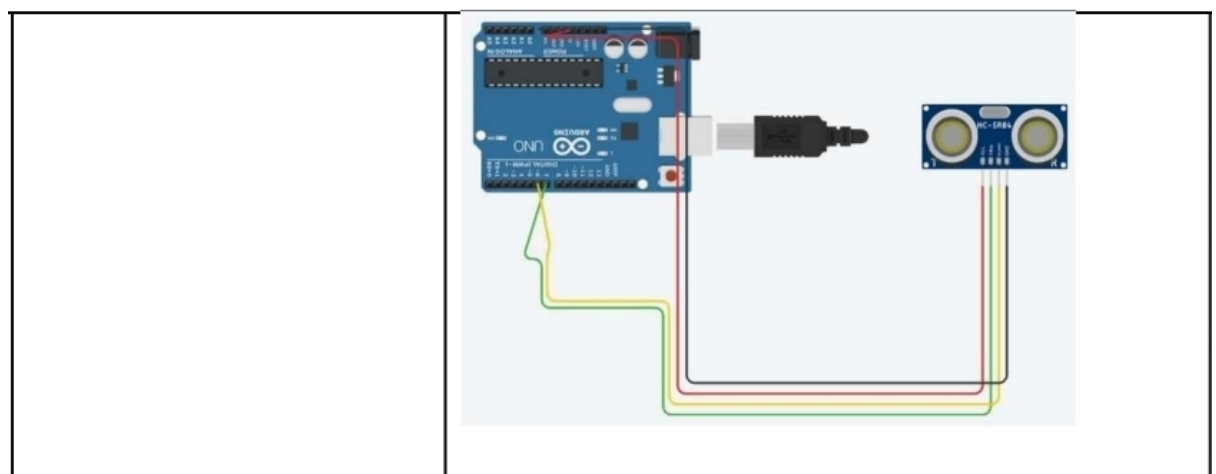
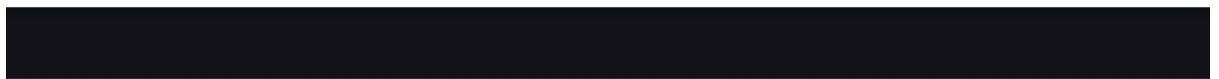
```

```
pinMode(pingPin, OUTPUT);  
digitalWrite(pingPin, LOW);  
delayMicroseconds(2);  
  
digitalWrite(pingPin, HIGH);  
delayMicroseconds(10);  
digitalWrite(pingPin, LOW);  
  
pinMode(echoPin, INPUT);  
duration = pulseIn(echoPin, HIGH);  
  
inches = duration / 74 / 2;  
cm = duration / 29 / 2;  
  
Serial.print(inches);  
Serial.print("in, ");  
Serial.print(cm);  
Serial.print("cm");  
Serial.println();  
  
delay(100);  
}
```

Connections:

1. Connect VCC → 5V on Arduino
2. Connect GND → GND
3. Connect TRIG → Digital Pin 7
4. Connect ECHO → Digital Pin 6

To interface Ultrasonic Sensor	Ultrasonic Sensor
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5. PIR SENSOR

```
int pirsensor = 0;
```

```
void setup() {
```

```
pinMode(2, INPUT);
pinMode(13, OUTPUT);
Serial.begin(9600);
}

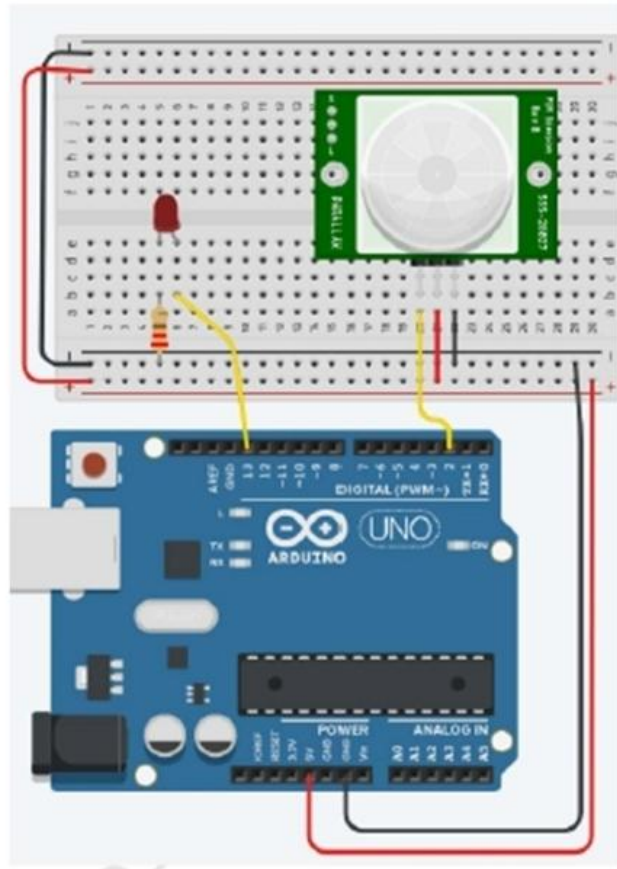
void loop() {
  pirsensor = digitalRead(2);

  if (pirsensor == 1) {
    digitalWrite(13, HIGH);
    Serial.println("Motion detected");
  } else {
    digitalWrite(13, LOW);
    Serial.println("Motion not detected");
  }

  delay(1000);
}
```

To interface PIR Sensor

PIR Sensor



Connections:

PIR Pin Connection

VCC 5V

GND GND

OUT Pin 2

LED:

Pin 13 → LED

6. DHT11 SENSOR

```
#include <dht.h>
```

```
dht DHT;
```

```
#define DHT11_DPIN 2
```

```
void setup() {  
  Serial.begin(9600);  
}
```

```

void loop() {
  int chk = DHT.read11(DHT11_DPIN);

  Serial.print("Temperature = ");
  Serial.println(DHT.temperature);

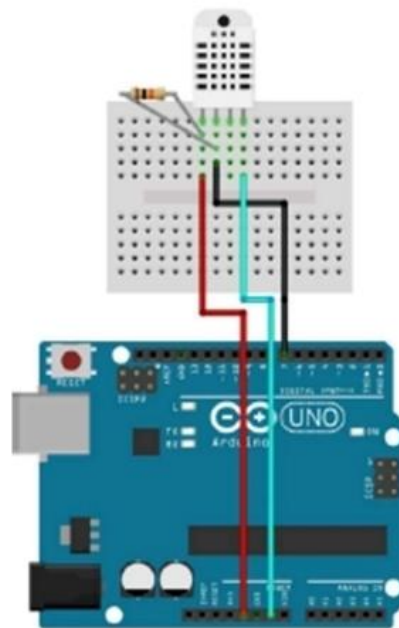
  Serial.print("Humidity = ");
  Serial.println(DHT.humidity);

  delay(1000);
}

```

To interface DHT11:
Humidity & Temperature
Sensor

DHT11: Humidity & Temperature Sensor



DHT11 Pin Connect To
VCC 5V
(Arduino)
GND GND
DATA Digital Pin 2

7. RAIN SENSOR

```

const int sensorMin = 0;

```



```
const int sensorMax = 1024;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int sensorReading = analogRead(A0);

  int range = map(sensorReading, sensorMin, sensorMax, 0, 3);

  switch (range) {
    case 0:
      Serial.println("RAINING");
      break;

    case 1:
      Serial.println("RAIN WARNING");
      break;

    case 2:
      Serial.println("NOT RAINING");
      break;
  }

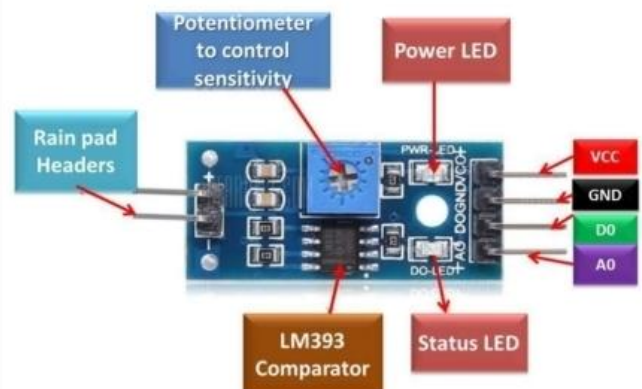
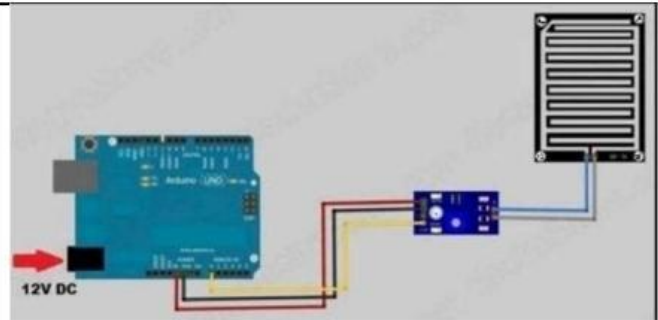
  delay(1000);
}
```

VCC → 5V

⌚ GND → GND

⌚ AO → A0

To interface Raindrop
Sensor



8. SOIL MOISTURE SENSOR

Analog Mode

```
int sensor_pin = A0;
int output_value;

void setup() {
  Serial.begin(9600);
}

void loop() {
  output_value = analogRead(sensor_pin);
  output_value = map(output_value, 550, 0, 0, 100);

  Serial.print("Moisture: ");
  Serial.print(output_value);
  Serial.println("%");

  delay(1000);
}
```

Digital Mode

```
int led_pin = 13;
int sensor_pin = 8;

void setup() {
  pinMode(led_pin, OUTPUT);
  pinMode(sensor_pin, INPUT);
}

void loop() {
  if (digitalRead(sensor_pin) == HIGH) {
    digitalWrite(led_pin, HIGH);
  } else {
    digitalWrite(led_pin, LOW);
  }

  delay(1000);
}
```

❑ VCC → 5V

❑ GND → GND

❑ AO → A0

9. LDR SENSOR

```
int sensorPin = A0;
int sensorValue = 0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  delay(100);
}
```

Component Connection

LDR one end 5V

LDR other end A0 +
resistor

Resistor other
end GND

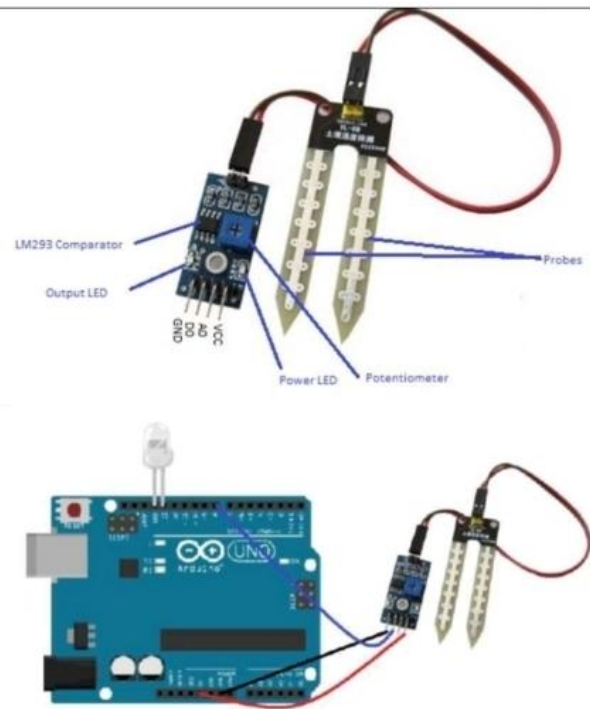
10. DC Motor

```
int motorpin1 = 6;
int motorpin2 = 7;

void setup() {
  pinMode(motorpin1, OUTPUT);
  pinMode(motorpin2, OUTPUT);
}

void loop() {
  digitalWrite(motorpin1, LOW);
  digitalWrite(motorpin2, HIGH);
}
```

To interface Soil
Moisture Sensor



11. LCD DISPLAY

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
void setup() {  
  lcd.begin(16, 2);  
}
```

```
void loop() {  
  lcd.setCursor(0, 0);  
  lcd.print("16x2 LCD MODULE");  
  
  lcd.setCursor(2, 1);  
  lcd.print("hello,world!");  
}
```

LCD

Pin Name Connect To

1 VSS GND

2 VDD 5V

3 V0 Middle pin of
potentiometer

4 RS Pin 12

5 RW GND

6 E Pin 11

11 D4 Pin 5

12 D5 Pin 4

13 D6 Pin 3

14 D7 Pin 2

15 A

(LED+) 5V (via 220Ω resistor)

16 K (LED-

) GND

Pot

Pin Connect To

Left GND

Right 5V

Middle LCD Pin 3

(V0)

12. DC MOTOR SPEED + DIRECTION

```
#define button 8
```

```
#define pot 0
```

```
#define pwm1 9
```

```
#define pwm2 10
```

```
boolean motor_dir = 0;
```

```
int motor_speed;
```

```
void setup() {
```

```
  pinMode(button, INPUT_PULLUP);
```

```
  pinMode(pwm1, OUTPUT);
```

```
  pinMode(pwm2, OUTPUT);
```

```
}
```

```
void loop() {
```

```
  motor_speed = analogRead(pot) / 4;
```

```
  if (motor_dir)
```

```
    analogWrite(pwm1, motor_speed);
```

```
  else
```

```
    analogWrite(pwm2, motor_speed);
```

```
  if (!digitalRead(button)) {
```

```
    while (!digitalRead(button));
```

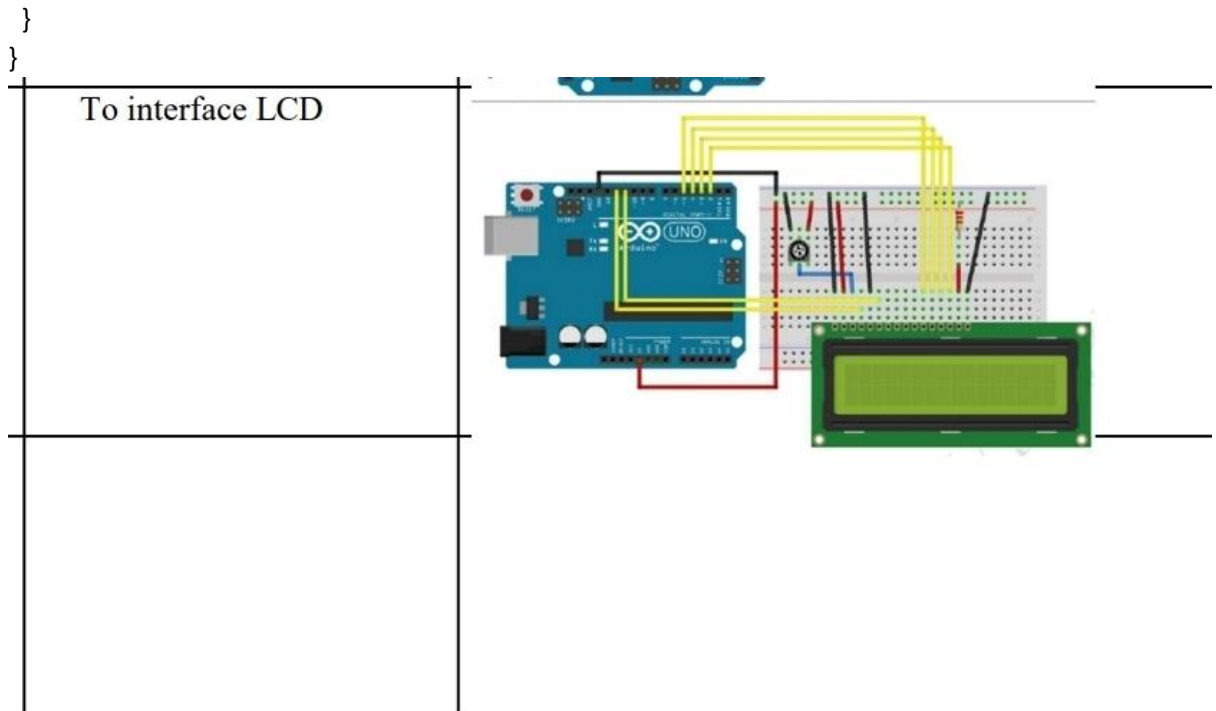
```
    motor_dir = !motor_dir;
```

```
  if (motor_dir)
```

```
    digitalWrite(pwm2, 0);
```

```
  else
```

```
    digitalWrite(pwm1, 0);
```



13. Sending sensor data to cloud using Raspberry- Pi(ESP8266)

```

#include <ESP8266WiFi.h>
#include "ThingSpeak.h"
#include "DHT.h"

#define DHTPIN D4
#define DHTTYPE DHT11
#define MQ3_PIN A0

const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

unsigned long channelID = YOUR_CHANNEL_ID;
const char* writeAPIKey = "YOUR_WRITE_API_KEY";

WiFiClient client;
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);
  dht.begin();

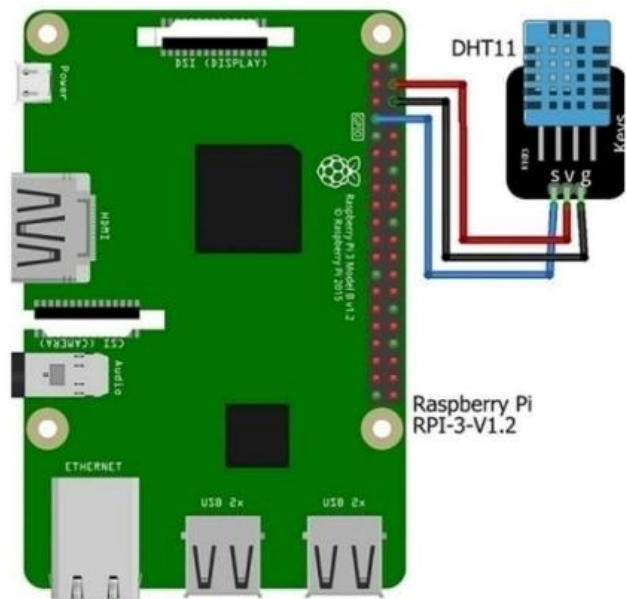
  WiFi.begin(ssid, password);
  Serial.print("Connecting");

```

```
while (WiFi.status() != WL_CONNECTED) {  
  delay(500);  
  Serial.print(".");  
}  
  
Serial.println("\nConnected to WiFi");  
ThingSpeak.begin(client);  
}  
  
void loop() {  
  float humidity = dht.readHumidity();  
  float temperature = dht.readTemperature();  
  int smokeValue = analogRead(MQ3_PIN);  
  
  Serial.print("Temperature: ");  
  Serial.println(temperature);  
  
  Serial.print("Humidity: ");  
  Serial.println(humidity);  
  
  Serial.print("Smoke Level: ");  
  Serial.println(smokeValue);  
  
  ThingSpeak.setField(1, temperature);  
  ThingSpeak.setField(2, humidity);  
  ThingSpeak.setField(3, smokeValue);  
  
  ThingSpeak.writeFields(channelID, writeAPIKey);  
  
  delay(15000); // ThingSpeak limit
```


}

Sending sensor data to
cloud using Raspberry- Pi



DHT11 Pin Connect To
VCC 3.3V
(ESP8266)
GND GND
DATA D4 (GPIO2)

14. ESP8266 WEB CONTROL (BULB)

```
#include <ESP8266WiFi.h>
```

```
const char* ssid = "";  
const char* password = "";
```

```
int ledPin = 13;  
WiFiServer server(80);
```

```
void setup() {  
  Serial.begin(115200);  
  pinMode(ledPin, OUTPUT);
```

```
  WiFi.begin(ssid, password);
```

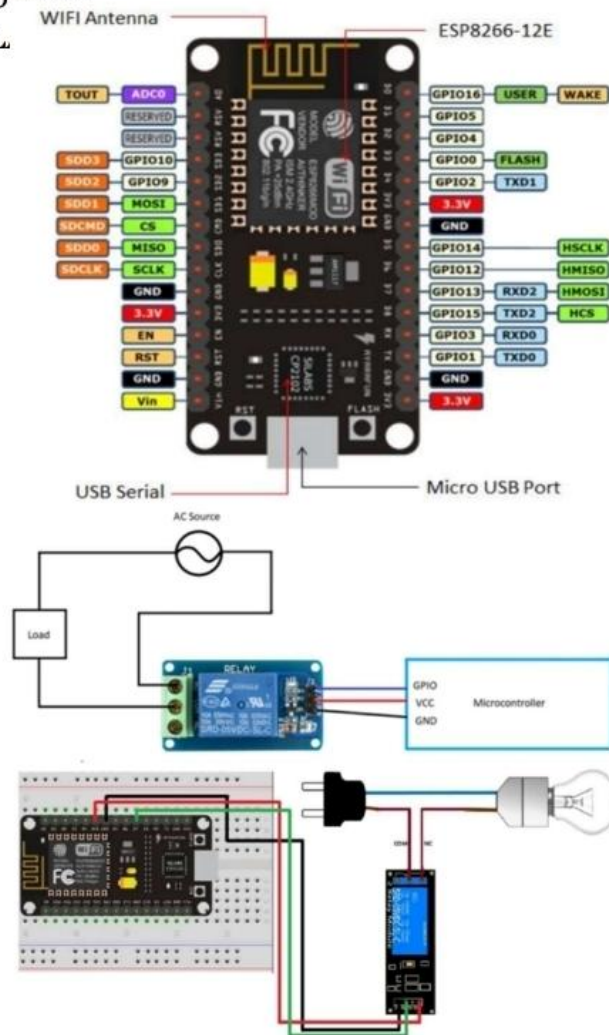
```
  while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
  }
```

```
  server.begin();  
}
```

```
void loop() {  
  WiFiClient client = server.available();  
  if (!client) return;  
  
  while (!client.available()) delay(1);  
  
  String request = client.readStringUntil('\r');  
  
  if (request.indexOf("/LED=ON") != -1) {  
    digitalWrite(ledPin, LOW);  
  }  
  
  if (request.indexOf("/LED=OFF") != -1) {  
    digitalWrite(ledPin, HIGH);  
  }  
}
```

Relay Pin Connect To
VCC 5V (or Vin)
GND GND
IN D7 (GPIO13)

Get the status of a bulb at a remote place (on the LAN) through web.



15.Home Automation

```
#include <Servo.h>
```

```
Servo door;
```

```
int pirPin = 2;
int trigPin = 10;
int echoPin = 11;
int motorEnable = 9;
int in1 = 8;
int in2 = 7;
int tempPin = A0;
```

```
int ledMotion = 3;
int ledTemp = 4;
```

```
void setup() {
  pinMode(pirPin, INPUT);
```

```

pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);

pinMode(in1, OUTPUT);
pinMode(in2, OUTPUT);
pinMode(motorEnable, OUTPUT);

pinMode(ledMotion, OUTPUT);
pinMode(ledTemp, OUTPUT);

door.attach(6);
Serial.begin(9600);
}

void loop() {

  // 🔥 PIR Sensor
  int motion = digitalRead(pirPin);
  digitalWrite(ledMotion, motion);

  // 📏 Ultrasonic
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;

  if (distance < 20) {
    door.write(90); // open
  } else {
    door.write(0); // close
  }

  // 🌡️ Temperature (LM35)
  int tempValue = analogRead(tempPin);
  float voltage = tempValue * (5.0 / 1023.0);
  float temperature = voltage * 100;

  if (temperature > 30) {
    digitalWrite(in1, HIGH);
    digitalWrite(in2, LOW);
    analogWrite(motorEnable, 200);
    digitalWrite(ledTemp, HIGH);
  } else {

```

```
digitalWrite(in1, LOW);  
digitalWrite(in2, LOW);  
digitalWrite(ledTemp, LOW);  
}  
Serial.print("Temp: ");  
Serial.println(temperature);  
delay(500);  
}
```

TINKERR CADD

DC MOTOR

<https://www.tinkercad.com/things/idZ5xxB4LV3-dc-motor-speed-and-direction?sharecode=dTnumsuT7Ds54OV -HvbMJScddS5WJROCwEE4JWeZ8Y>

AUTOMATION

<https://www.tinkercad.com/things/g4qhAFNDekk-swanky-blad?sharecode=TsYekjdjqOcxQIN6HP0GeNrKPqeQiGSvrQ4epneH0-k>